PORTFOLIO MANAGEMENT & CRISIS BACKTESTING

SAPM | IMT Ghaziabad
Historical portfolio optimisation

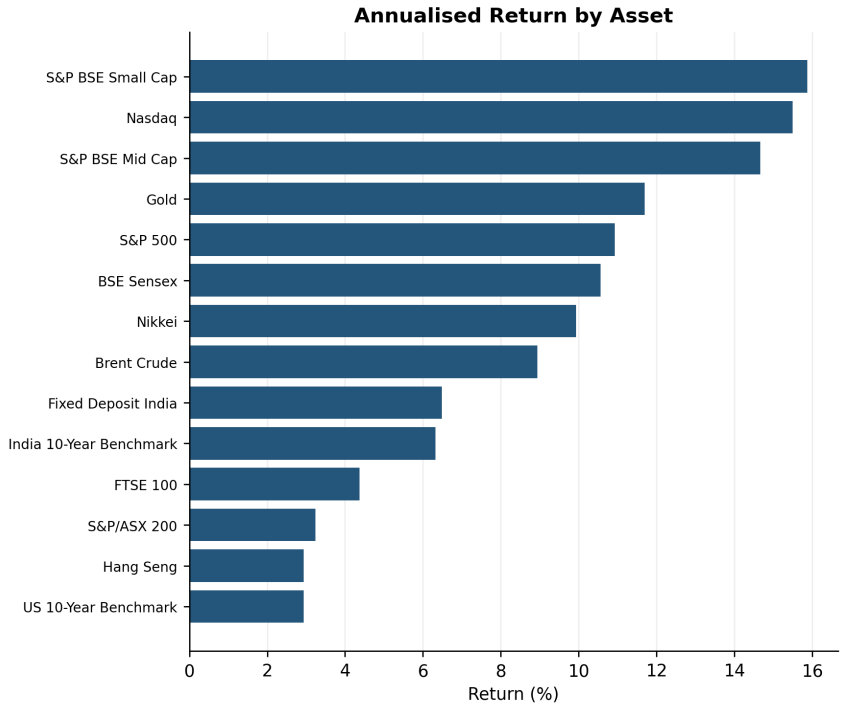
01 | Optimal Portfolio Allocation

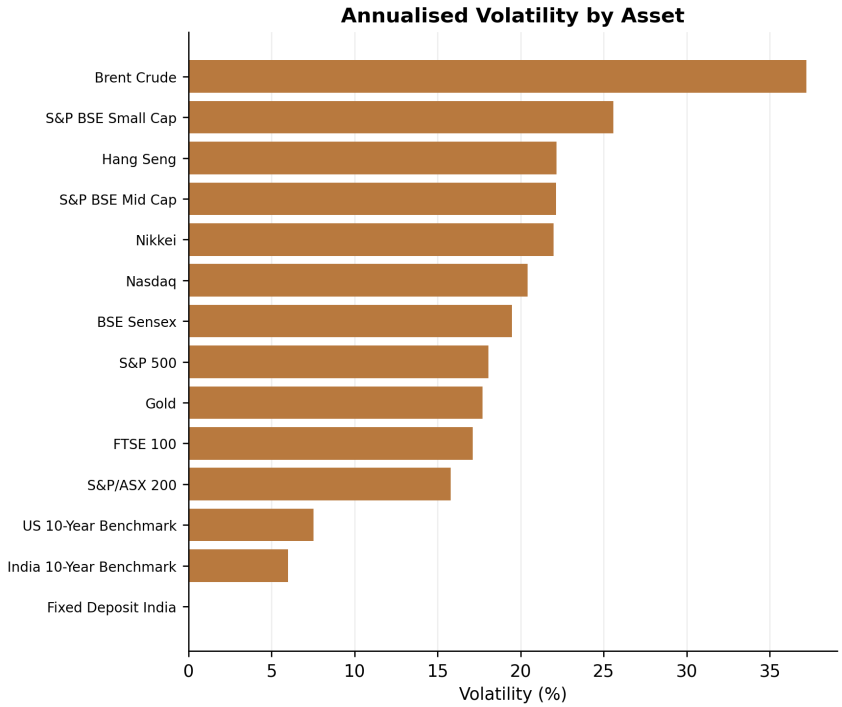
Asset	Weight (%)
S&P BSE Mid Cap	24.87
Nasdaq	21.01
S&P BSE Small Cap	20.99
Gold	19.39
Fixed Deposit India	13.74

02 | Portfolio Performance

Metric	Value
Annual Return	0.13
Annual Volatility	0.11
Sharpe Ratio	0.60
Number of Assets	5.00

03 | Asset-Level Risk and Return


Annualised Return by Asset
S&P BSE Small Cap
Nasdaq
S&P BSE Mid Cap
Gold
S&P 500
BSE Sensex
Nikkei
Brent Crude
Fixed Deposit India
India 10-Year Benchmark
FTSE 100
S&P/ASX 200
Hang Seng
US 10-Year Benchmark
0
2
4
6
8
10
12
14
16
Return (%)


Annualised Volatility by Asset
Brent Crude
S&P BSE Small Cap
Hang Seng
S&P BSE Mid Cap
Nikkei
Nasdaq
BSE Sensex
S&P 500
Gold
FTSE 100
S&P/ASX 200
US 10-Year Benchmark
India 10-Year Benchmark
Fixed Deposit India
0
5
10
15
20
25
30
35
Volatility (%)

Crisis-Period Performance

Unnamed: 0	2008 GFC	2020 COVID	2022 Russia-Ukraine
Q1 Portfolio Return (%)	-41.43	28.78	-6.09
Annualized Volatility (%)	20.08	16.54	10.87

Subsequent-Period Performance

Unnamed: 0	2008 GFC Stress Portfolio	2020 COVID Stress Portfolio	2022 Ukraine Stress Portfolio
Subsequent Period Return (%)	32.43	16.78	27.70
Annualized Volatility (%)	7.32	11.35	11.38

05 | Stress Portfolio Comparison

index	Q1 Optimal Portfolio	2008 GFC Stress Portfolio	2020 COVID Stress Portfolio	2022 Russia-Ukraine Stress Portfolio
Annual Return (%)	12.54	1.18	23.86	4.37
Annual Volatility (%)	10.99	11.34	19.86	14.93
Sharpe Ratio	0.60	-0.42	0.90	-0.11

Methodology and Limitations

The initial portfolio was selected using a maximum-Sharpe-ratio approach subject to allocation constraints. Stress portfolios optimised using returns from the same crisis year are ex-post historical stress tests and involve hindsight. Currency effects are ignored as instructed. The fixed-deposit return and risk-free rate are project assumptions. Historical performance is not a forecast.